Supplementary Material:
A Scene-Aware Bidirectional Evaluation on RDD2022

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Scope and conventions

This supplement accompanies the three-variant, 18-run evaluation. All results are calculated from the supplied archive metrics. Percentages are AP or mAP multiplied by 100; gaps are in percentage points. Uncertainty denotes sample standard deviation over seeds 13, 37, and 71. No confidence interval, additional training run, or independent inference replay is introduced. CSV files preserve unrounded values.

Complete class-level results

Table S1 gives the four-class AP matrix. Rows sharing a target use identical test images. Means are not rankings of individual runs. YOLO26n has the highest in-domain mean mAP but does not have the highest AP for every class.

Table S1: Full-test AP@50:95 by class (%, three-seed means). Class-specific SDs are retained in `data/metric_summary.csv`.

Model	Train	Target	D00	D10	D20	D40
YOLOv8n	Drone	Drone	30.96	34.66	19.45	0.57
YOLOv8n	MotorBike	Drone	6.27	8.91	0.72	10.31
YOLO11n	Drone	Drone	32.45	36.48	17.42	3.20
YOLO11n	MotorBike	Drone	6.51	11.26	1.43	9.80
YOLO26n	Drone	Drone	33.37	39.57	22.94	2.06
YOLO26n	MotorBike	Drone	10.07	13.34	2.89	4.02
YOLOv8n	Drone	MotorBike	20.58	16.23	2.14	1.11
YOLOv8n	MotorBike	MotorBike	47.18	50.66	55.35	51.83
YOLO11n	Drone	MotorBike	20.98	17.35	1.00	1.26
YOLO11n	MotorBike	MotorBike	48.31	53.05	58.72	52.27
YOLO26n	Drone	MotorBike	24.27	22.75	1.54	1.38
YOLO26n	MotorBike	MotorBike	50.44	53.66	57.99	56.93

Seed consistency and complementary metric

Each entry in Table S2 subtracts the other-domain-trained score from the target-domain-trained score on identical target images for one model and seed. All 18 full-test gaps are positive. Gap SDs in the main paper are calculated from these differences before aggregation. Table S3 provides full-test mAP@50.

Table S2: Full-test fixed-target mAP@50:95 gaps (pp) by seed.

Model	Target	13	37	71
YOLOv8n	Drone	15.03	13.52	16.01
YOLOv8n	MotorBike	40.39	42.40	40.93
YOLO11n	Drone	16.27	10.84	18.29
YOLO11n	MotorBike	43.49	42.47	42.85
YOLO26n	Drone	16.91	16.10	17.71
YOLO26n	MotorBike	43.14	42.87	40.80

Table S3: Full-test mAP@50 (%, mean $\pm$ sample SD).

Model	Train	Drone	MotorBike
YOLOv8n	Drone	44.17 ± 2.77	24.29 ± 3.28
YOLOv8n	MotorBike	18.02 ± 6.99	84.68 ± 0.13
YOLO11n	Drone	45.51 ± 3.73	23.44 ± 2.34
YOLO11n	MotorBike	20.81 ± 11.78	84.54 ± 0.31
YOLO26n	Drone	48.73 ± 2.09	27.47 ± 0.44
YOLO26n	MotorBike	18.11 ± 0.61	86.75 ± 0.74

Excluded-label sensitivity and validation dynamics

The clean-only rule removes images with excluded-class annotations, leaving 273 Drone and 271 MotorBike images; it does not require all four target classes in each image. Table S4 reports the scores. All 18 clean-only seed gaps are positive. Mean ± SD gaps on Drone/MotorBike targets are $15.25 \pm 1.58 / 41.87 \pm 1.34$ pp for YOLOv8n, $14.93 \pm 4.42 / 43.86 \pm 0.63$ pp for YOLO11n, and $16.59 \pm 0.95 / 42.60 \pm 1.87$ pp for YOLO26n. Score changes cannot be isolated from the changed image population.

Table S4: Clean-only mAP@50:95 (%, mean ± sample SD).

Model	Train	Drone	MotorBike
YOLOv8n	Drone	23.03 ± 1.22	10.33 ± 1.35
YOLOv8n	MotorBike	7.78 ± 2.73	52.20 ± 0.46
YOLO11n	Drone	23.71 ± 2.50	10.72 ± 0.58
YOLO11n	MotorBike	8.79 ± 4.05	54.59 ± 0.60
YOLO26n	Drone	26.09 ± 0.91	13.57 ± 0.12
YOLO26n	MotorBike	9.49 ± 0.47	56.17 ± 1.76

Table S5: Epoch of maximum logged source-validation mAP@50:95.

Model	Train	Seed 13	Seed 37	Seed 71
YOLOv8n	Drone	63	133	129
YOLOv8n	MotorBike	120	109	119
YOLO11n	Drone	110	92	111
YOLO11n	MotorBike	128	131	113
YOLO26n	Drone	117	81	119
YOLO26n	MotorBike	126	82	128

Fig. S1 displays all 18 source-validation trajectories; Table S5 records their mAP-maximum epochs. Every run completed 150 epochs with patience zero. A logged validation maximum is distinct from independently verifying the checkpoint-selection fitness. These curves contain no test-set metrics.

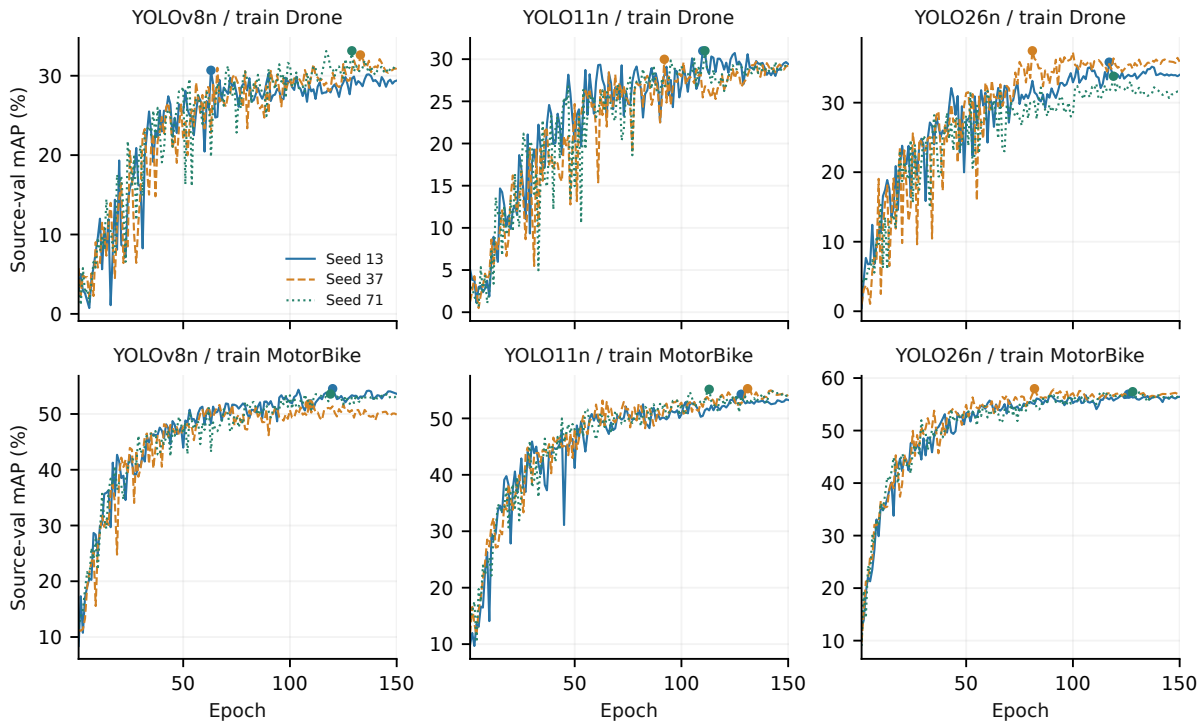


Figure S1: Source-validation mAP@50:95 for three YOLO variants and two training domains. Each panel contains seeds 13, 37, and 71; circles mark their maximum logged validation mAP. Line styles distinguish seeds as well as colors.

Regeneration and evidence boundary

The package retains the 72 metric rows, run identities, 18 training-log CSVs, and summary tables. Model, training domain, target domain, seed, and subset jointly identify a metric row; `D13` alone is not unique. The supplied preparation and build scripts regenerate tables, figures, and PDFs from retained records; commands are documented in the package README. The optional white-background gap-summary figure is supplied separately, because the same numbers already appear in the main paper. Regeneration reproduces manuscript artifacts from archived summaries, not training or original-image evaluation.